

Data Warehouse & Business Intelligence Lab

Instructor: Dr. Khurram Shahzad

BS Data Science Fall 24-M

Lab 02 | 16-09-2026

Objective: *This lab aims to give students hands-on practice with database case studies by helping them distinguish between transactional and analytical queries. Students will also learn to apply conditional statements and decoding functions, work with various date formats, and address challenges related to multivalued and composite attributes in database design.*

Case Study (PUCIT Data Science Department)

Case Study Background

The Data Science Department at Punjab University College of Information Technology (PUCIT) maintains student, teacher, course, and enrollment data. When preparing this data for analysis, the department faces recurring issues:

1. **Performance Classification:** Student marks are stored as raw numbers, but decision-making often requires converting them into categories (Pass/Fail, Grades, Scholarship Eligibility). Analysts must learn to **use conditional logic** to derive these attributes directly from data.
2. **Date Issues:** Admission and registration dates are stored in a single numeric string format (e.g., 16092025 for 16-Sep-2025). Some records contain **invalid or illogical values** (e.g., day 00 or 32, month > 12, February 31). Analysts need to **reformat valid dates** and **identify invalid ones** for cleaning and reporting.
3. **Multivalued & Composite Attributes:**
 - Some students have **multiple phone numbers** stored in one field.
 - Full names are stored as one string, making it harder to extract **first name and last name**.
 - Students enroll in multiple courses, and teachers teach multiple courses, requiring aggregation in analysis.

The department needs analysts to write SQL queries to resolve these problems so the data can be used for reliable reports and dashboards.

Teacher Assistants: Khadija Shahzad, Rida Batool, Areeba Noor

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Tasks

Part A – Conditional Statements & Decoding Functions

1. Display all students with their marks in “Database Management Systems,” and show “Pass” if marks ≥ 50 , otherwise “Fail.”
2. Count the number of *Pass* and *Fail* students in each course of the Data Science department.
3. Create grade categories (A for ≥ 85 , B for 70–84, C for 55–69, D for 40–54, F for < 40) and display them with student names and course names.
4. Show students eligible for scholarships. Criteria: average marks ≥ 80 across all their enrolled courses.

Part B – Working with Date Formats

5. Retrieve admission dates and convert them to DD/MM/YYYY format using string functions.
6. Retrieve registration dates and display them as YYYY-MM-DD.
7. Identify students who registered in the year 2023 by extracting the year portion from RegistrationDate.
8. List all students with **invalid admission dates** (e.g., day=00 or >31, month=00 or >12, Feb 30/31)

Part C – Handling Multivalued & Composite Attributes

9. Extract the **primary phone number** (first number) from the PhoneNumbers column.
10. Split student FullName into **FirstName** and **LastName**.
11. Generate a report showing **each student** with a **comma-separated list of courses** they are enrolled in.
12. Generate a report showing **each teacher** in Data Science with the number of courses they teach.

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